A projective plane of order 18 admits no collineation of order 17 (computer-assisted, conditional on assumptions A1–A5)

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Abstract

We prove that a projective plane of order 18 has no collineation σ of order 17. Counting arguments and Bruck's bound on subplanes (only $m^2 \le n$ is used) show that σ has either exactly three fixed points, forming a triangle, or exactly twenty, namely a line plus an external point. The second case is excluded by an orbit-matrix argument over $\mathbb{Z}[\zeta_{17}]$: since 2 has order 8 modulo 17, the primes above 2 are self-conjugate, which forces the number of non-fixed orbits to be even, but it is 19. For the first case we derive from the axioms a reduction to 58,161 equivalence classes of orthomorphisms of $\mathbb{Z}_{17}$ with pairwise compatibility conditions on "rows"; an exhaustive computer search shows that no class has a solution. The triangle case is therefore computer-assisted, and the theorem holds under assumptions A1–A5 (Section 7.3). Baer's theorem on involutions is not used.

1 Introduction

A projective plane of order n has $n^2 + n + 1$ points, $n + 1$ points on every line, and a unique line through any two points. Planes of prime-power order exist; none is known otherwise, and the Bruck–Ryser theorem [4] excludes infinitely many orders. Order 10 was excluded by a large computation [7] building on coding theory [8]. Order $18 = 3^2 + 3^2$ is not excluded by Bruck–Ryser.

Theorem 1.1. *A projective plane Π of order 18 has no collineation of order 17. This is computer-assisted: the case of three fixed points is proved by an exhaustive search, and the theorem holds under the assumptions A1–A5 of Section 7.3.*

Context. Baer [2, Thm. 5, Cor. 6] shows that in a plane of non-square order every involution is a perspectivity; we do not use this. MacWilliams–Sloane–Thompson [8] (text not obtained; we rely on the report in [7, Sec. 2]), the survey [6], and [1, 9] concern order 10 and codes of designs. Janko–Tran Van Trung [5] (text not obtained, not relied on) restrict collineation groups for small orders. Orbit matrices over group rings are a standard tool; we found no primary reference stating the norm-parity lemma of Section 4 in the form used here and make no claim that it is new. We were *unable to verify* whether Theorem 1.1 or the exclusion of the homology configuration already appears in the literature (e.g. for orders $12, 15, 18$ or orbit matrices of $(343, 19, 1)$ designs); the elementary parts may be routine to experts. We do not claim that Theorem 1.1 is new; we were unable to find it in the literature we could access. The reduction of the triangle case to orthomorphisms of $\mathbb{Z}_{17}$ and its complete enumeration may be of interest.

2 Preliminaries

Throughout, Π has order $n = 18$, $v = n^2 + n + 1 = 343$, and σ is a collineation of prime order $p = 17$; orbit indices live in $\mathbb{Z}_{17} = \mathbb{Z}/17$. Quoted facts (nothing else from the literature is used):

Q1. (Counting.) The incidence matrix N satisfies $NN^T = N^T N = nI + J$.

Q2. (Bruck [3, Lemma 3.1].) If a plane of order n has a proper subplane of order m, then $n = m^2$ or $n \ge m^2 + m$; we use only $m^2 \le n$.

Q3. (Cyclotomic fields.) For $\zeta = e^{2\pi i/p}$, the ring of integers of $K = \mathbb{Q}(\zeta)$ is $\mathbb{Z}[\zeta]$, $\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/p)^*$ via $\gamma_a : \zeta \mapsto \zeta^a$, and γ_{-1} is complex conjugation. A prime $q \neq p$ is unramified, and the decomposition group of each prime above q is generated by γ_q.

Lemma 2.1. *Let σ have prime order p in a plane of order n, with f fixed points.*

(a) Orbits on points and on lines have length 1 or p; $f \equiv n^2 + n + 1 \pmod p$, and a fixed line carrying f_L fixed points has $f_L \equiv n + 1 \pmod p$.

(b) The numbers of fixed points and of fixed lines are equal; hence so are the numbers of non-fixed point orbits and line orbits.

(c) The line through two fixed points is fixed; the intersection of two fixed lines is fixed.

Proof. (a) Orbit lengths divide p. Non-fixed points form orbits of length p; so do the non-fixed points of a fixed line, which has $n + 1$ points. (b) Let A, B be the permutation matrices of σ on points and lines; $AN = NB$. By Q1, $\det N \neq 0$ ($\det(nI + J) = n^{v-1}(n + v)$), so $B = N^{-1}AN$ and $\operatorname{tr} A = \operatorname{tr} B$, these being the numbers of fixed points and lines. (c) σ maps the line xy (x, y fixed) to a line through x, y; dually. □

Here $343 \equiv 3$ and $19 \equiv 2 \pmod{17}$.

3 The fixed structure

Let $F = \mathrm{Fix}(\sigma)$, $f = |F|$. A *quadrangle* is a set of four points, no three collinear.

Lemma 3.1. *If F contains a quadrangle, then F is the point set of a proper subplane of order $m \geq 2$, and $f = m^2 + m + 1$.*

Proof. Let $\mathcal{L}_F$ be the fixed lines. By Lemma 2.1(c), two points of F lie on a unique line, which is in $\mathcal{L}_F$, and two lines of $\mathcal{L}_F$ meet in a point of F. The quadrangle makes the structure non-degenerate, so $(F, \mathcal{L}_F)$ is a projective plane of some order $m \geq 2$. If $m = n$ then F is everything and $\sigma = \mathrm{id}$; so $m < n$. □

Lemma 3.2. *If F contains no quadrangle, there are a line ℓ and a point P with $F \subseteq \ell \cup \{P\}$.*

Proof. If F is collinear, take $\ell \supseteq F$. Otherwise take non-collinear $A, B, C \in F$ with *sides* AB, AC, BC, and $X = F \setminus \{A, B, C\}$. *Step 1: each $D \in X$ lies on a side.* Otherwise no three of A, B, C, D are collinear, so $\{A, B, C, D\}$ is a quadrangle. *Step 2.* Two sides meet in a vertex, which is not in X; so by Step 1 the parts of X on the three sides partition X. *Step 3: X meets at most one side.* Let two sides be UV, UW (common vertex U), and suppose $D \in X \cap UV$, $E \in X \cap UW$; then $D, E \neq U$. We claim $\{V, W, D, E\}$ is a quadrangle: V, W, D collinear would put W on $VD = UV$ (but $W \notin UV$); V, W, E collinear would put V on $WE = UW$ (but $V \notin UW$); V, D, E collinear would put E on $VD = UV$, hence $E \in UV \cap UW = \{U\}$; W, D, E collinear would put $D = U$. All are impossible, a contradiction. Hence X lies on one side, say AB, and $F \subseteq AB \cup \{C\}$. □

Theorem 3.3 (Fixed structure)**.** *Let σ have order 17 in a plane of order 18. Then $f \in \{3, 20\}$, and*

(i) if $f = 3$, the fixed points P_0, P_1, P_2 form a triangle, the fixed lines are exactly its three sides, and each side carries exactly two fixed points;

(ii) if $f = 20$, $F = \ell \cup \{P\}$ with $P \notin \ell$, and there are exactly 20 fixed lines.

The number of non-fixed point orbits (equal to that of line orbits) is $r = (343 - f)/17$, i.e. $r = 20$ for $f = 3$ and $r = 19$ for $f = 20$.

Proof. By Lemma 2.1(a), $f \equiv 3 \pmod{17}$. If F contains a quadrangle, Lemma 3.1 and Q2 give $m^2 \leq 18$, $m \in \{2, 3, 4\}$, $f \in \{7, 13, 21\}$, none $\equiv 3$. So Lemma 3.2 applies and $f \leq 19 + 1 = 20$; thus $f \in \{3, 20\}$. If $f = 20$, then $F = \ell \cup \{P\}$ with 20 distinct points, so $P \notin \ell$, and there are 20

fixed lines by Lemma 2.1(b). If $f = 3$ and the fixed points lay on a line L, L would be fixed with $f_L = 3 \not\equiv 2$, contradicting Lemma 2.1(a). So they form a triangle; its sides are fixed, and there are exactly three fixed lines by (b). □

4 The homology configuration is impossible

Theorem 4.1 (Norm-parity lemma)**.** *Let σ have odd prime order p in a plane of order n, with r orbits of length p on points. Let $q \neq p$ be a prime with $-1 \in \langle q \bmod p \rangle \leq (\mathbb{Z}/p)^*$. Then $r\, v_q(n)$ is even.*

Proof. Let $\zeta = e^{2\pi i/p}$. Choose representatives $x_1, \dots, x_r$ of the point orbits and $y_1, \dots, y_r$ of the line orbits of length p (equal numbers by Lemma 2.1(b)), and set

$$A_{ab} = \sum_{k \in \mathbb{Z}_p} \zeta^k N[x_a, \sigma^k y_b] \in \mathbb{Z}[\zeta].$$

Put $u_a = \sum_k \zeta^k e_{\sigma^k x_a}$ and $w_b = \sum_k \zeta^k e_{\sigma^k y_b}$ (standard Hermitian form); they have disjoint supports, $\langle u_a, u_{a'} \rangle = p\delta_{aa'}$, $\langle w_b, w_{b'} \rangle = p\delta_{bb'}$. *Claim:* $Nw_b = \sum_a A_{ab} u_a$. At $x = \sigma^m x_a$: $(Nw_b)(x) = \sum_k \zeta^k N[x_a, \sigma^{k-m} y_b] = \zeta^m A_{ab} = A_{ab} u_a(x)$. At a fixed point x: $(Nw_b)(x) = N[x, y_b] \sum_k \zeta^k = 0$. Since $\sum_k \zeta^k = 0$, $Jw_{b'} = 0$, so by Q1 $\langle Nw_b, Nw_{b'} \rangle = \langle w_b, N^T N w_{b'} \rangle = n \langle w_b, w_{b'} \rangle = np\, \delta_{bb'}$. By the claim the left side is $p \sum_a A_{ab} \overline{A_{ab'}}$. Hence $A^* A = nI_r$ and $\det A \cdot \overline{\det A} = n^r$. Let $\mathfrak{p} \mid q$ be a prime of $\mathbb{Z}[\zeta]$. By Q3 its decomposition group $\langle \gamma_q \rangle$ contains γ_{-1}, so complex conjugation fixes $\mathfrak{p}$ and $v_\mathfrak{p}(\overline{\alpha}) = v_\mathfrak{p}(\alpha)$. As q is unramified, $v_\mathfrak{p}(n) = v_q(n)$, and applying $v_\mathfrak{p}$ gives $2 v_\mathfrak{p}(\det A) = r\, v_q(n)$. □

Corollary 4.2. *There is no collineation of order 17 with $f = 20$ in a plane of order 18.*

Proof. Take $p = 17$, $q = 2$: $2^4 = 16 \equiv -1$, so 2 has order 8 and $-1 \in \langle 2 \rangle$; $v_2(18) = 1$. So r is even, but $r = 19$ by Theorem 3.3. □

Remark 4.3. For $f = 3$, $r = 20$ is even and nothing follows; for $q = 3$, $v_3(18) = 2$ and the condition is vacuous. $\mathbb{Q}(\zeta_{17})$ has $16/8 = 2$ primes above 2, each fixed by complex conjugation.

5 The triangle case: reduction to a finite problem

Now σ has exactly three fixed points P_0, P_1, P_2. Write $\{a, b, k\} = \{0, 1, 2\}$ and $L_k = P_a P_b$. Only $p = 17$ prime and $n = p + 1$ are used, so the derivation can be tested with other p (Section 7). The letters r_i, c_j, τ_k below are gauge parameters.

Lemma 5.1 (Orbit structure)**.** *A point is* general *if on no side; a line is* general *if not a side and containing no P_k.*

(a) $S_k := L_k \setminus \{P_a, P_b\}$ is one orbit of length 17.

(b) The set V_k of the 17 lines through P_k other than the sides L_a, L_b is one orbit. A line of V_k meets L_k in one point (of S_k), meets L_a, L_b only in P_k, and its other 17 points are general.

(c) There are 289 general points, forming orbits $O_1, \dots, O_{17}$, and 289 general lines, forming orbits $H_1, \dots, H_{17}$.

(d) A general line meets each L_k in one point of S_k, these three points are distinct, and it has 16 general points. A general point lies on exactly one line of each V_0, V_1, V_2 and on 16 general lines.

(e) Every line of V_k contains exactly one point of each O_i.

Proof. (a) L_k has 19 points, two fixed; the other 17 are non-fixed, hence one orbit. (b) Of the 19 lines through P_k, two are sides; the other 17 are not fixed (the fixed lines are the sides), so form one orbit. Such a line meets L_k outside P_a, P_b (else it is a side $P_k P_a$ or $P_k P_b$), and meets L_a, L_b only

in P_k; of its 19 points one is P_k, one is on L_k, and 17 are general. (c) $343-3-3\cdot 17=289$ general points, non-fixed, so 17 orbits. Lines containing a vertex or being a side number $3+3\cdot 17=54$ (a non-side contains at most one vertex), leaving 289 non-fixed lines. (d) A general line Λ meets each side in a non-vertex point, so in S_k; two of these points coincide only at a vertex, so they are distinct and Λ has $19-3=16$ general points. For a general point Q, the lines QP_k are distinct non-sides, so $QP_k\in V_k$ (the three lines QP_0, QP_1, QP_2 are distinct, else Q lies on a side), and the other 16 lines through Q are general. (e) By (d), $\sum_{v\in V_k}|v\cap O_i|=17$; as V_k is one orbit and O_i is invariant, $|v\cap O_i|=c_i$ is constant, so $c_i=1$. □

5.1 Coordinates and incidence identities

Choose $v_k\in V_k$, $s_k:=v_k\cap L_k$, and put $V_{k,w}=\sigma^w v_k$, $(k,z)=\sigma^z s_k$ $(w,z\in\mathbb{Z}_{17})$, so $V_{k,w}\cap L_k=\{(k,w)\}$. Choose $Q_i\in O_i$, $\Lambda_j\in H_j$ $(i,j\in[17]:=\{1,\dots,17\})$, and put $(i,X)=\sigma^X Q_i$, $H_{j,y}=\sigma^y\Lambda_j$. Define

$$f_{ij}=\{t: Q_i\in\sigma^t\Lambda_j\},\quad s_j^k=\text{the } t \text{ with } s_k\in\sigma^t\Lambda_j,\quad u_i^k=\text{the } t \text{ with } Q_i\in\sigma^t v_k.$$

(Well defined: $\Lambda_j\cap L_k$ is one point of S_k and Q_i is on one line of V_k, by Lemma 5.1(d), and σ acts regularly on S_k and V_k.) Equivariance gives, for all i,j,k and $X,y,z,w\in\mathbb{Z}_{17}$:

$$(i,X)\in H_{j,y}\iff y-X\in f_{ij}, \tag{I1}$$
$$(k,z)\in H_{j,y}\iff y-z=s_j^k, \tag{I2}$$
$$(i,X)\in V_{k,w}\iff w-X=u_i^k, \tag{I3}$$
$$(k,z)\in V_{k,w}\iff z=w. \tag{I4}$$

By Lemma 5.1(b), $V_{k,w}$ contains no point of $S_{k'}$ $(k'\neq k)$; general lines contain no vertex, and sides no general point.

Proposition 5.2 (Incidence identities)**.** *(I) For each i, the multiset $\{t-t': j\in[17],\ t,t'\in f_{ij},\ t\neq t'\}$ equals $\mathbb{Z}_{17}^*$, each element once.*

(II) For each i,k, the sets $\{t-s_j^k: t\in f_{ij}\}$ $(j\in[17])$ are pairwise disjoint with union $\mathbb{Z}_{17}\setminus\{u_i^k\}$.

(III) For each j,k, the sets $\{u_i^k-t: t\in f_{ij}\}$ $(i\in[17])$ are pairwise disjoint with union $\mathbb{Z}_{17}\setminus\{-s_j^k\}$.

(IV) For $a\neq b$, $j\mapsto s_j^a-s_j^b$ is a bijection $[17]\to\mathbb{Z}_{17}$.

(V) For $i\neq i'$ put $\delta_k=u_i^k-u_{i'}^k$. Then $\delta_0,\delta_1,\delta_2$ are distinct, and $\{y-y': j,\ y\in f_{ij},\ y'\in f_{i'j}\}$ consists of the 14 elements of $\mathbb{Z}_{17}\setminus\{\delta_0,\delta_1,\delta_2\}$, each once.

(VI) For $j\neq j'$, the elements $s_{j'}^k-s_j^k$ $(k=0,1,2)$ are distinct, and $\{t'-t: i,\ t\in f_{ij},\ t'\in f_{ij'}\}$ consists of the 14 elements outside them, each once. For $j=j'$, $\{t'-t: i,\ t\neq t'\in f_{ij}\}=\mathbb{Z}_{17}^*$, each once.

Proof. Each item counts the unique line through two points (or the unique common point of two lines), and decides to which family that line belongs.

(I) For $d\neq 0$ the points $(i,0),(i,d)$ lie on one line λ, not a side, and not in V_k (Lemma 5.1(e)); so $\lambda=H_{j,y}$ for a unique (j,y), and by (I1) $y,\ y-d\in f_{ij}$. Conversely every such triple gives a line through both points.

(II) For $z\in\mathbb{Z}_{17}$, the line through $(i,0)$ and (k,z) is no side and not in $V_{k'}$, $k'\neq k$. It lies in V_k iff it is $V_{k,w}$ with $w=z$ (I4) and $w=u_i^k$ (I3), i.e. iff $z=u_i^k$. Otherwise it is $H_{j,y}$ with $y\in f_{ij}$ (I1) and $y-z=s_j^k$ (I2), i.e. $z=t-s_j^k$, $t=y\in f_{ij}$, for a unique (j,t).

(III) $H_{j,0}$ and $V_{k,w}$ meet in one non-vertex point Z. If Z is a side point it is (k,w), and lies on $H_{j,0}$ iff $w=-s_j^k$ (I2). If $Z=(i,X)$ is general, (I3) gives $X=w-u_i^k$ and (I1) gives $-X\in f_{ij}$, i.e. $w=u_i^k-t$, $t=-X\in f_{ij}$.

(IV) (a,z) and (b,z') lie on one line, not a side and not in any V_c (a line of V_c has exactly one point on $S_0 \cup S_1 \cup S_2$). So it is $H_{j,y}$ with $y - z = s_j^a$, $y - z' = s_j^b$, i.e. $z' - z = s_j^a - s_j^b$, for a unique j.

(V) $(i,0)$ and (i',X) lie on one line, not a side. It is in V_k iff it is $V_{k,w}$ with $w = u_i^k$ and $w - X = u_{i'}^k$, i.e. $X = \delta_k$. If $\delta_k = \delta_{k'}$ for $k \neq k'$, two distinct lines would contain the same two points; so the δ_k are distinct. For $X \notin \{\delta_k\}$ the line is $H_{j,y}$ with $y \in f_{ij}$, $y - X \in f_{i'j}$, for a unique (j,y).

(VI) For $(j',y) \neq (j,0)$, $H_{j,0}$ and $H_{j',y}$ meet in one non-vertex point. It is a side point (k,z) iff $-z = s_j^k$ and $y - z = s_{j'}^k$ (I2), i.e. iff $y = s_{j'}^k - s_j^k$. It is general (i,X) iff $t := -X \in f_{ij}$, $t' := y - X \in f_{ij'}$ (I1), i.e. $y = t' - t$. If $j = j'$, $y \neq 0$, a side point would need $y = 0$, so the point is general. If $j \neq j'$, the three $s_{j'}^k - s_j^k$ are distinct (else the lines share two side points), and the count follows. □

Types of rows. By (I), a block f_{ij} of size m contributes $\binom{m}{2}$ of the 8 classes $\{\pm d\}$, each exactly once over j. So $\sum_j \binom{|f_{ij}|}{2} = 8$, whence $|f_{ij}| \leq 4$, and $\sum_j |f_{ij}| = 16$ by (II). With $\nu_m = \#\{j : |f_{ij}| = m\}$ (not to be confused with the order n): $6\nu_4 + 3\nu_3 + \nu_2 = 8$, $4\nu_4 + 3\nu_3 + 2\nu_2 + \nu_1 = 16$, with solutions $(\nu_4,\nu_3,\nu_2,\nu_1) \in \{(0,0,8,0),(0,1,5,3),(0,2,2,6),(1,0,2,8)\}$ ($\nu_4 \leq 1$; $\nu_4 = 1$ forces $\nu_3 = 0, \nu_2 = 2$; if $\nu_4 = 0$, $\nu_3 \leq 2$).

5.2 Gauge, normal form, nets

Replacing $Q_i \mapsto \sigma^{r_i} Q_i$, $\Lambda_j \mapsto \sigma^{c_j}\Lambda_j$, $v_k \mapsto \sigma^{\tau_k} v_k$ changes the data to

$$f'_{ij} = f_{ij} + r_i - c_j, \qquad s'^k_j = s^k_j - c_j + \tau_k, \qquad u'^k_i = u^k_i + r_i - \tau_k$$

(e.g. $Q'_i \in \sigma^t \Lambda'_j \iff Q_i \in \sigma^{t+c_j-r_i}\Lambda_j$).

Proposition 5.3 (Normal form)**.** *The choices can be made with $s_j^0 = 0$ and $u_i^0 = 0$ for all i,j. Then (a) for each i the sets f_{ij} ($j \in [17]$) partition $\mathbb{Z}_{17}^*$, and for each j the sets f_{ij} ($i \in [17]$) partition $\mathbb{Z}_{17}^*$; (b) $j \mapsto s_j^1$, $j \mapsto s_j^2$, $j \mapsto s_j^2 - s_j^1$ are bijections $[17] \to \mathbb{Z}_{17}$.*

Proof. Take $\tau_k = 0$, $c_j = s_j^0$, $r_i = -u_i^0$. (a) is Proposition 5.2(II),(III) for $k = 0$ (in (III) the sets $-f_{ij}$ partition $\mathbb{Z}_{17} \setminus \{0\}$). (b) is (IV) applied to the three pairs of sides, using $s^0 = 0$. □

The residual freedom is $c_j = r_i = \tau_0$ constant and arbitrary τ_1, τ_2: all f_{ij} are unchanged and $s_j^k \mapsto s_j^k + \tau_k - \tau_0$ ($k = 1,2$).

The net. By (b) name the columns by $\alpha = s_j^1 \in \mathbb{Z}_{17}$, write $f_{i\alpha}$, and define $\theta(\alpha) = s_j^2$. Then θ and $\theta - \mathrm{id}$ are bijections: θ is an *orthomorphism* of $\mathbb{Z}_{17}$, and $\Theta = \{(s_j^1, s_j^2)\}$ is a transversal of the cyclic Latin square of order 17 (it meets each of $x = c$, $y = c$, $y - x = c$ once, as θ and $\theta - \mathrm{id}$ are bijections). Residual translations replace Θ by $\Theta + (a,b)$, so we may assume $\theta(0) = 0$; let Orth_0 be the set of such θ.

5.3 Rows and compatibility

For each row i define $g_i : \mathbb{Z}_{17}^* \to \mathbb{Z}_{17}$ by $g_i(x) = \alpha$ iff $x \in f_{i\alpha}$ (Proposition 5.3(a)); the *cells* are $C_\alpha = g^{-1}(\alpha)$.

Definition 5.4 (Rows)**.** $\mathrm{Rows}(\theta)$ is the set of $g : \mathbb{Z}_{17}^* \to \mathbb{Z}_{17}$ such that: (R1) the multiset of differences $x - x'$ ($x \neq x'$ in a common cell) is exactly $\mathbb{Z}_{17}^*$; (R2) $x \mapsto x - g(x)$ is injective; $u^1(g)$ denotes the missing element of $\mathbb{Z}_{17}$; (R3) $x \mapsto x - \theta(g(x))$ is injective; $u^2(g)$ denotes the missing element; (R4) for columns $\alpha \neq \beta$ the differences $y - x$ ($x \in C_\alpha, y \in C_\beta$) are pairwise distinct and differ from $\beta - \alpha$ and $\theta(\beta) - \theta(\alpha)$.

Proposition 5.5 (Rows)**.** *In the normal form each $g_i \in \mathrm{Rows}(\theta)$, with $u^k(g_i) = u_i^k$ ($k = 1,2$).*

For a fixed net the rows are independent objects: $\mathrm{Rows}(\theta)$ depends on θ only, and the only coupling is pairwise compatibility.

Proof. (R1) is (I). (R2), (R3) are (II) for $k = 1, 2$ with $s_j^1 = \alpha_j$, $s_j^2 = \theta(\alpha_j)$. (R4): for columns $\alpha \neq \beta$ ($j \neq j'$), by (VI) the differences $t' - t$ over all rows are distinct and avoid $s_{j'}^k - s_j^k$, i.e. $\beta - \alpha$ and $\theta(\beta) - \theta(\alpha)$; restrict to row i. □

Definition 5.6 (Compatibility)**.** Rows $g, g' \in \mathrm{Rows}(\theta)$ are *compatible* if, with $\delta_0 = 0$, $\delta_k = u^k(g) - u^k(g')$: (C1) $\delta_0, \delta_1, \delta_2$ are distinct and $\{y - y' : \alpha \in \mathbb{Z}_{17},\ y \in C_\alpha,\ y' \in C'_\alpha\}$ consists of the 14 elements of $\mathbb{Z}_{17} \setminus \{\delta_0, \delta_1, \delta_2\}$, each once; (C2) for columns $\alpha \neq \beta$ the sets $\{y - x : x \in C_\alpha, y \in C_\beta\}$ and $\{y - x : x \in C'_\alpha, y \in C'_\beta\}$ are disjoint.

Proposition 5.7 (Pairs)**.** *In the normal form, $g_i, g_{i'}$ are compatible for $i \neq i'$.*

Proof. (C1) is (V) (with $\delta_0 = u_i^0 - u_{i'}^0 = 0$); (C2) is (VI), since all differences over all rows in a column pair are distinct. □

Remark 5.8. Compatible rows satisfy $\sum_\alpha |C_\alpha||C'_\alpha| = 14$ and $g(x) \neq g'(x)$ for all x (since $0 \notin$ (C1)), and no two difference classes $\pm d$ occur once in a cell of g and once in a cell of g' of the same column. The program uses these as pre-filters; they add nothing logically.

Lemma 5.9 (Bucket lemma)**.** *In the normal form, for each $x \in \mathbb{Z}_{17}^*$ the map $i \mapsto g_i(x)$ is a bijection $[17] \to \mathbb{Z}_{17}$. Hence with $B_\theta(x, c) = \{g \in \mathrm{Rows}(\theta) : g(x) = c\}$ (the* buckets*), if some $g_i \in B_\theta(1, 0)$, then for every $c \neq 0$ the bucket $B_\theta(1, c)$ contains a row compatible with g_i (namely the $g_{i'}$ with $g_{i'}(1) = c$).*

Proof. By Proposition 5.3(a), for each column α the cells $f_{i\alpha}$ ($i \in [17]$) partition $\mathbb{Z}_{17}^*$, so x lies in exactly one. □

5.4 Symmetry and classes

Let G be the group generated by translations of $\mathbb{Z}_{17}^2$, the maps $(x, y) \mapsto (mx, my)$, $m \in \mathbb{Z}_{17}^*$, and $(x, y) \mapsto (y, x)$, $(x, y) \mapsto (-x, y - x)$ (these two generate S_3).

Proposition 5.10 (Symmetry)**.** *If (Π, σ) of triangle type yields the net Θ in normal form, then for every $\gamma \in G$ there are coordinates of the same type yielding $\gamma(\Theta)$.*

Proof. Translations: the residual freedom. Scalars: replace σ by the generator σ^m; then s_j^k becomes $m^{-1}s_j^k$. Relabel the vertices by a permutation π: $s_j^k \mapsto s_j^{\pi(k)}$, and renormalise by subtracting $s_j^{\pi(0)}$; the new net is $\{(s_j^{\pi(1)} - s_j^{\pi(0)},\ s_j^{\pi(2)} - s_j^{\pi(0)})\}$. In the coordinates $(x, y) = (s_j^1 - s_j^0, s_j^2 - s_j^0)$, swapping $1, 2$ is $(x, y) \mapsto (y, x)$ and swapping $0, 1$ is $(x, y) \mapsto (-x, y - x)$. □

Orthomorphisms $\theta, \theta' \in \mathrm{Orth}_0$ are *equivalent* if $\gamma(\mathrm{graph}\,\theta) = \mathrm{graph}\,\theta'$ for some $\gamma \in G$. A class consists of the normalised orthomorphisms obtained from $L(\mathrm{graph}\,\theta)$, for L in the group of $16 \cdot 6 = 96$ linear maps, by translating one of its 17 points to the origin; so it has at most $17 \cdot 96 = 1632$ elements. Its *representative* is the lexicographically smallest $(\theta(1), \ldots, \theta(16))$.

Theorem 5.11 (Reduction)**.** *If Π has order 18 and a collineation σ of order 17 with exactly 3 fixed points, there are a class representative θ and pairwise compatible rows $g_1, \ldots, g_{17} \in \mathrm{Rows}(\theta)$ such that $g_1(x), \ldots, g_{17}(x)$ are distinct for every x. In particular, for the row g with $g(1) = 0$ and every $c \in \{1, \ldots, 16\}$, $B_\theta(1, c)$ contains a row compatible with g.*

Proof. Normal form (Proposition 5.3) with θ a representative (Proposition 5.10); rows by Proposition 5.5, compatibility by Proposition 5.7, distinctness and the last claim by Lemma 5.9. □

All conditions used are necessary; compatibility is weaker than the full set of incidence identities, which suffices since only non-existence is to be shown. The search therefore tests, for each class representative θ,

$$\forall g \in B_\theta(1, 0)\ \ \exists c \in \{1, \ldots, 16\} :\ \text{no row of } B_\theta(1, c) \text{ is compatible with } g. \qquad (\star_\theta)$$

6 The computation

The 1,606,008,513 orthomorphisms of $\mathbb{Z}_{17}$, of which 94,471,089 are normalised, fall into 58,161 classes under G (exhaustive enumeration with a canonicity test, `orth.cpp`); Table 1 gives the class sizes. The set $\mathcal{F}$ of partitions of $\mathbb{Z}_{17}^*$ satisfying (R1) ("families") has $490{,}801 = 3857 + 351360 + 119168 + 16416$ members, for the four types $(\nu_4, \nu_3, \nu_2, \nu_1)$ of Section 5, in the order listed there (obtained by enumeration; no closed formula is known; the count is used only to describe the search).

class size	3	6	102	204	408	816	1632	total
number of classes	1	2	3	6	27	486	57 636	58 161
orthomorphisms	3	12	306	1 224	11 016	396 576	94 061 952	94 471 089

Table 1: Classes of normalised orthomorphisms of $\mathbb{Z}_{17}$. The sizes $3, 6, 6$ account for the 15 linear orthomorphisms $\theta(x) = \lambda x$, $\lambda \neq 0, 1$, on which S_3 acts anharmonically (a consistency check).

Algorithm 1 Test of $(\star_\theta)$ for one representative θ

Require: $\theta \in \mathrm{Orth}_0$; the list $\mathcal{F}$ (same for all θ)

1: generate lazily $B(c) = B_\theta(1, c)$, $c = 0, \ldots, 16$: place the blocks of each family in distinct columns with 1 in column c; prune by (R2),(R3); check (R4); record u^1, u^2
2: $\mathit{Cand} \leftarrow B(0)$
3: **for** $c = 1, \ldots, 16$ **while** $\mathit{Cand} \neq \emptyset$ **do**
4: $\quad \mathit{Cand} \leftarrow \{g \in \mathit{Cand} : \exists h \in B(c)$ compatible with $g\}$ ▷ necessary by Lemma 5.9
5: **end for**
6: **return** ELIMINATED if $\mathit{Cand} = \emptyset$; else run an exact 17-clique search with one row from each bucket containing a survivor

Compatibility is tested in the order: (a) $\sum_\alpha |C_\alpha||C'_\alpha| = 14$ (SIMD); (b) $g(x) \neq h(x)$ for all x; (c) no common same-column difference class; (d) (C1) by bitmasks; (e) (C2) by bitsets indexed by (column pair, difference). Tests (a)–(c) are implied by (d),(e) (Remark 5.8).

Theorem 6.1 (Computational)**.** $(\star_\theta)$ *holds for each of the* 58,161 *class representatives. Indeed* $\mathit{Cand} = \emptyset$ *after the step* $c = 3$ *for every* θ*; the exact search is never invoked (zero survivors,* 17*-cliques and nodes).*

Established by computer; data in `run_out.txt` (one line per net), summarised in Table 2.

quantity	value
class representatives processed	58 161 (ids 0–58160, once each)
$\lvert B(0)\rvert$, range (mean)	23 696–27 776 (26 774)
nets with $\mathit{Cand} = \emptyset$ first at $c = 1$ / $c = 2$ / $c = 3$	0 / 57 538 / 623
survivors after $c = 1$: total (max per net)	971 489 (37)
survivors after $c = 2$: total (max per net)	627 (2)
survivors after $c = 3$; 17-cliques; exact-search nodes	0; 0; 0
total CPU time	179 471 s ($\approx$ 49.9 core-hours)

Table 2: The run for $p = 17$, recomputed from `run_out.txt` by one `awk` line.

Proof of Theorem 1.1. By Theorem 3.3, $f \in \{3, 20\}$. Corollary 4.2 excludes 20. If $f = 3$, Theorem 5.11 gives a representative θ for which $(\star_\theta)$ fails, contradicting Theorem 6.1. □

7 Verification, assumptions and limitations

7.1 Controls

The reduction uses only that σ has prime order p, $n = p+1$, and a fixed triangle. Replacing 17 by p throughout ($(p-1)/2$ difference classes, $p-3$ for 14, etc.), the same program (`-DP=`p) applies to $n = p+1$.

p	n	$\lvert\mathrm{Orth}_0\rvert$	classes	with sol.	plane of order n	output
3	4	1†	1	1	exists	as expected (1 clique)
5	6	3	1	0	none	as expected
7	8	19	3	2	exists	as expected
11	12	3 441	11	0	open	no information
13	14	79 259	117	0	none	as expected
31	32	—	—	—	PG(2,32)	true plane found (below)
17	18	94 471 089	58 161	0	open	—

Table 3: Controls. $p = 7$: 4 cliques on the 3 representatives; on all 133 transversal nets (no reduction) 35 nets carry 70 solutions, each verified to be a plane of order 8 (`plane.py`). $p = 5$: the family list has one member and no valid row exists. †`orth.cpp` needs $p \geq 5$; for $p = 3$ the unique $\theta(x) = 2x$ was entered by hand. Entries with output 0 are weak tests (a bug discarding too much also gives 0); the positive controls are $p = 3, 7$, and $p = 31$ only via ported predicates (see below).

Why positive controls are in characteristic 2. A genuine control needs a plane of order $n = p + 1$ with a collineation of prime order p with a fixed triangle. All known planes have prime-power order q^k with $q^k - 1$ prime: for odd q this forces $n = 3$, $p = 2$ (excluded, p must be odd); for $q = 2$, $p = 2^k - 1$ is a Mersenne prime, $n = 4, 8, 32, 128, \ldots$. We use the Desarguesian planes $\mathrm{PG}(2, 2^k)$ with $2^k - 1$ prime as controls; they have such a collineation (diag of order $2^k - 1$). Other known planes with such an automorphism may exist (e.g. non-Desarguesian planes of order 32) but were not used. A non-Desarguesian plane of order q^2, $q > 2$, has $n - 1 = (q - 1)(q + 1)$ composite. So the controls available are $p = 3, 7, 31, 127$; $p = 127$ was not attempted. The $p = 3, 7$ controls test that true solutions survive the necessary conditions, pruning and the bucket lemma, but not features specific to 17: the family list, blocks of size 4 (which need $p \geq 13$), or the code paths for 17 columns.

Control $p = 31$ ($\mathrm{PG}(2, 32)$). Elements $\sigma_c = \mathrm{diag}(1, g, g^c)$, $c = 2, \ldots, 30$ (for $c = 2$ alone all blocks have size ≤ 2, as for $p = 7$; for σ_c blocks are roots of a trinomial, with sizes $1, 2, 3, 5$; size 4 occurs only at $N = 63$, composite). From the explicit geometry of the 29 elements, the quotient data were computed, and the normal form, (R1)–(R3), the net conditions, the bucket argument and the compatibility predicates accepted all: 31/31 true rows, 465/465 true row pairs; about 5000 single-entry perturbations of true rows rejected; the exact search finds exactly one 31-clique, the true plane; the 1057-point structure rebuilt from the quotient data satisfies the axioms. These tests used hand-ported 64-bit predicates and generators (`ctrl_p.cpp`; regression on $\mathrm{PG}(2, 8), \mathrm{PG}(2, 16)$ passes). A parametrised copy of `fast4.cpp` reproduces the first 500 lines of the $P = 17$ run and the $P = 7$ results on all 133 nets, and at $P = 31$ returns exactly the true plane for $c = 3, \ldots, 30$ except 16, and 6 variants, with and without decoy (extra random false) families; at $P = 63$ blocks of size 4 were tested with injected rows (generator bypassed). *Not done:* the original `fast4.cpp` at $p = 31$ (its SIMD tables are fixed at 17); $c = 2$, $c = 16$ and the all-pairs row type (generator memory); the full $\mathrm{Rows}(\theta)$ at $p = 31$; the generator on families with 4-blocks; equality of the parametrised build with the $p = 17$ binary on all 58,161 nets. The control shows the quotient derivation is not contradicted by real planes; it says nothing about non-existence.

7.2 Independent checks

- **Independent re-implementations from the same derivation.** `indep.cpp` agrees with the main program byte for byte on the row sets $B(1,0), B(1,1), B(1,2)$ of a test net and on survivor counts. `mine.cpp` ($p = 3, 5, 7, 11, 13$) and `mine17b.cpp` ($p = 17$) agree on all controls and, for $p = 17$, on $|B(1,0)|, a_1, a_2, a_3$ for 40 random nets and 8 nets with $a_2 > 0$ (a_i: candidates left after filtering step i).
- **Reproduction from this paper alone.** A second implementation, written without the original code, reproduced the 490,801 families, the 94,471,089 orthomorphisms and the 58,161 classes with the same size histogram (its own canonical form), passed the controls $p = 3, 5, 7, 11, 13$ (these controls were run by the independent reproducer; we cannot confirm a re-run of the original implementation when preparing this text), and decided all 58,161 classes with 0 cliques (interrupted once at 8,926 classes, resumed from checkpoint). Verdicts agree; per-class statistics are not comparable.
- **Counting.** The class sizes sum to 94,471,089; `cnt.cpp` (depth-first) gives the same and $\times 17 =$ 1,606,008,513; 200 random orthomorphisms canonicalise into the list; `orbsum.cpp` and `orb.cpp` enumerate orbits explicitly; the class list regenerated from scratch has the same sorted content. The family count and its split were reproduced by `fam.cpp` and `indep.cpp`.
- **Oracle and SAT.** `oracle_pairs.py` tests compatibility of 257 pairs on a net by building the partial incidence structure from the axioms; it agrees with the C++ function. A direct CP-SAT model of the axioms with only the gauge conditions $s^0 = u^0 = 0$ confirms Proposition 5.3 for $p = 3, 7$ (infeasible when negated) and shows infeasibility for $p = 5$.
- **Re-run.** For this paper 2,760 nets of the $p = 17$ run (ids 0–759, 30000–31999) were re-run with output identical to `run_out.txt` except run time. The full $p = 17$ search was not repeated.

7.3 Assumptions

Theorem 1.1 rests on:

A1. the quoted facts Q1–Q3 (Bruck's bound is the only non-elementary plane-geometric input);

A2. the correctness of the hand derivation of Section 5 (re-derived by a second, non-blind reader and consistent with the CP-SAT model for $p = 3, 7$; no human referee);

A3. the correctness of the enumeration of class representatives and of Proposition 5.10;

A4. the correctness of the search with respect to Definitions 5.4 and 5.6: its row generator produces all of $B_\theta(1, c)$, and its compatibility test implies (C1),(C2). Only necessary conditions are tested, so a test that is too weak cannot lose a solution, whereas one that is too strong could;

A5. correct execution on the hardware used.

7.4 Limitations

This is a computer-assisted result with no human referee. The row generator (forward checking, trie of block structures, SIMD) is checked against a naive generator and independent implementations on some nets but is not formally verified; a proof-producing check (an independent simple program recomputing $B(1, c)$, or a certificate for "no compatible row") would be valuable. All implementations, including the reproduction from the paper, share the derivation of Section 5: they are independent in implementation, not in logic, and the logic was re-derived only by a second non-blind reader. Proposition 5.10 was also checked numerically ($p = 7, 11$: row counts and solution counts are constant on classes). At $p = 31$ the production code was ported and run, but the full row set was not enumerated, $c = 2, 16$ and `a28_b25` were not run, and the 4-block generator was not tested; this control is partial, and a derivation-independent check is open. Novelty is not verified. The result concerns only the prime 17; the primes $2, 3, 5, 7, 19$ remain open for order 18, and existence of a plane of order 18 is untouched.

AI assistance. Most derivations, programs and verification were produced by AI agents (large language models, Claude by Anthropic) under the author's direction; the author set the problem, directed the research and judged the results. The argument does not rely on their authority: every step of Sections 3–4 and 5 is written out for human checking, and the code and data are held by the author (Section A). The work has not been independently re-checked by a human specialist; a human re-derivation of Section 5 and a run on other hardware are the most useful checks.

A Reproduction

The programs are in the directory `code/` of the repository (`fast4.cpp`, `orth.cpp`, and the 133 nets of the $p = 7$ control, `nets7_all.txt`; MIT licence). The auxiliary programs named in the verification section (independent re-implementations, counting, the $p = 31$ control, the geometric oracle, `plane.py`, `watchdog.sh`; the reproduction from the paper alone is `indep2.cpp`) are in `code/verification/`, as they were used and without polishing; the `README.md` there says which program backs which claim. The output of the original run is not distributed: the computation is deterministic, so a new run reproduces it (the time column excepted). Requirements: AVX2, OpenMP; `-march=native` makes the build machine dependent. Compiler: MinGW-w64 GCC 16.1.0 (WinLibs); the compiler of the original run was not recorded and is assumed the same.

```
g++ -O3 -march=native -fopenmp orth.cpp -o orth
./orth classes classes17.txt      # 58161 lines
g++ -O3 -march=native -fopenmp -DP=17 fast4.cpp -o netrun17.exe
OMP_NUM_THREADS=12 ./netrun17.exe classes17.txt 0 58161 run_out.txt run
awk '{a2+=$7;s+=$13;c+=$17} END{print NR,a2,s,c}' run_out.txt  # 58161 627 0 0
```

Interrupted runs resume by re-running the command (ids already in the output are skipped). The class list is written by `./orth classes <file>` (about 10 s; the order of the lines depends on thread scheduling, the set does not, so compare sorted files). Cost: about 3 s CPU per net on average (maximum 6.75 s), about 4 h wall clock on 12 threads; memory a few hundred MB per thread.

Interrupted run. The original run was started as `fastrun.exe`; after 727 nets the executable vanished from disk and the process ended (cause unknown; anti-virus suspected). The same source, compiled as `netrun17.exe` (byte-identical to `fast4.cpp`), was restarted by `watchdog.sh` (log: `done=727`) and finished in about 4 h. The output has 58,161 lines, ids $0, \ldots, 58160$ once each; the first 727 come from the first executable, whose exact command line (notes: `-O3 -march=native -DP=17`) was not preserved. These 727 nets were recomputed with identical results.

Checksums (SHA-256).

```
fast4.cpp      eb3f3ca800d6cee24255b14f25e543c7a30495051ed2f62e650ac112b37713bb
classes17.txt  aacf1791e82d5ad9b69a17ec32ef0d4c803a9a58e75585554bcbdcc72e8d7bbc
run_out.txt    86cff376cd5a3468aac2403618a6a046b7d260670ee347e85510b3f9f8598f1b
orth.cpp       74665a45165fc4c430778c9ade9b72209c521a073be2e8a9140c4792ab0af7a5
orbsum.cpp     3a96c58b848891994825974e3ca895e893dfb3484784a57164fd472ff24189ab
```

The files `classes17.txt` and `run_out.txt` of the author's run are not distributed; their checksums are listed for the record (the time column of `run_out.txt` changes from run to run). The file `orth.cpp` was edited after the first checksum was taken (output handling only); the hash above is that of the archived file. The sorted regenerated class list has SHA-256 `c3e3937b...e111e`, equal to that of the archived list sorted likewise.

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